

Lipase

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Diagnose pancreatitis, more specific for pancreatitis than is serum amylase; diagnose peritonitis, strangulated or infarcted bowel, pancreatic cyst

Contraindications

Urine specimens are inappropriate for lipase. Lipase activity is usually absent in urine, possibly from inactivation of the enzyme.

Custom Additional Information

Serum lipase is usually normal in patients with elevated serum amylase, without pancreatitis, who have peptic ulcer, salivary adenitis, inflammatory bowel disease, intestinal obstruction, and macroamylasemia. Coexistence of increased serum amylase with normal lipase may be a helpful clue to the presence of macroamylasemia.¹ Lipase is elevated with amylase in acute pancreatitis, but the elevation of lipase is more prolonged.

In work-up of pancreatitis, in addition to serum lipase and amylase, the 2-hour urine amylase is of value. Electrolytes, serum calcium, glucose, and acetone are also often needed. Immunoreactive trypsin is technically more difficult than lipase and probably no better.² The serum lipase:amylase ratio may help distinguish alcoholic from nonalcoholic pancreatitis. Ratios >2 (expressed as multiples of the upper limits of normal) suggest an alcoholic etiology.³ Lipase isoform or isoenzymes have been studied.⁴

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Range (U/L)
Male	
0 to 6 m	8–37
7 m to 1 y	11–34
2 to 17 y	11–38
≥18 y	13–78
Female	
0 to 6 m	9–50
7 m to 1 y	10–37
2 to 17 y	12–45
18 to 70 y	14–72
≥71 y	14–85

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Improper labeling

Footnotes

1. Andrews PA, Thomas PA. Macro-amylasaemia as a cause of persistently raised serum amylase. *Br J Surg*. 1988 Oct;75(10):1035. PubMed 2464398 (<http://www.ncbi.nlm.nih.gov/pubmed/2464398>)

2. Lott JA, Speicher CE, Nemesánszky E. Is serum amylase an obsolete test in the diagnosis of acute pancreatitis? *Arch Pathol Lab Med.* 1985 Apr;109(4):314-315. PubMed 2580500 (<http://www.ncbi.nlm.nih.gov/pubmed/2580500>)
3. Gumaste VV, Dave PB, Weissman D, Messer J. Lipase:amylase ratio. A new index that distinguishes acute episodes of alcoholic from nonalcoholic acute pancreatitis. *Gastroenterology.* 1991 Nov;101(5):1361-1366. PubMed 1718808 (<http://www.ncbi.nlm.nih.gov/pubmed/1718808>)
4. Lott JA, Lu CJ. Lipase isoforms and amylase isoenzymes: Assays and application in the diagnosis of acute pancreatitis. *Clin Chem.* 1991 Mar;37(3):361-368. PubMed 1706232 (<http://www.ncbi.nlm.nih.gov/pubmed/1706232>)